

Kevin DiPerna

Week Four - Assignment Centrality Measures

Data 620

1. Data Source

For this assignment, I selected the **GitHub Social Network** dataset from the Stanford Network Analysis Project (SNAP). The dataset was collected from GitHub using the public GitHub API and represents a network of software developers.

In this network, each node represents a GitHub developer, while edges represent mutual follower relationships between developers. The dataset contains approximately 37,700 developers and 289,000 edges, making it large enough to support meaningful centrality analysis.

An important feature of this dataset is that each developer also has a categorical label identifying the type of developer. The primary category identifies users as either **web developers** or **machine learning developers**. This makes the dataset appropriate for comparing network centrality measures across different groups.

2. Proposed Method for Loading the Data

If I were conducting the analysis, I would first download the edge list and node attribute data from the SNAP website. The data could then be loaded into Python using the pandas library for tabular data and NetworkX for network analysis.

The edge list would be used to construct the graph. Each developer would become a node, and each mutual follower relationship would become an edge between two nodes.

The node attribute file would then be merged with the network data so that each developer's categorical label could be included in the analysis. This would make it possible to calculate centrality measures for each developer and compare those values across the two developer categories.

3. Centrality Measures

The main centrality measure I would examine is **degree centrality**. Degree centrality measures how many direct connections a node has relative to the other nodes in the network.

In this dataset, a developer with high degree centrality would have mutual follower relationships with many other developers. These users could potentially occupy more prominent positions within the GitHub developer network.

Other centrality measures could also be useful, such as **betweenness centrality** and **closeness centrality**. Betweenness centrality could help identify developers who connect otherwise separate groups, while closeness centrality could show which developers are positioned closest to the rest of the network.

4. Comparison Across Categorical Groups

The categorical variable allows the centrality measures of **web developers** and **machine learning developers** to be compared.

For example, I could calculate the average degree centrality for each group and determine whether one type of developer tends to have more network connections than the other.

The analysis could also examine the distribution of degree centrality within each category rather than only comparing the averages. This would help determine whether differences are consistent across the groups or driven by only a small number of highly connected users.

5. Hypothetical Outcome

One possible outcome to predict would be whether developers with higher degree centrality are more likely to become influential participants in the GitHub community.

A possible hypothesis would be that developers with higher degree centrality have greater exposure to other developers and therefore may be more likely to participate in widely used open-source projects or become important members of collaborative development communities.

The analysis could also test whether developer category is associated with this outcome. For example, machine learning developers may have a different degree-centrality distribution than web developers because their communities, technologies, and collaboration patterns may differ.

Degree centrality alone would not fully measure developer influence, but it could serve as one useful predictor when combined with additional network measures and node attributes.

6. Data Source

Stanford Network Analysis Project (SNAP). **GitHub Social Network Dataset**.

Source: SNAP GitHub Social Network dataset.